PRE-ALGEBRA (8TH) · UNIT 7: GEOMETRY

Skate Park Design

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design a skate park: right triangles size the ramps and rails, the distance formula lays out paths on the map, and volume formulas produce a concrete order.

The Pythagorean Theorem, its converse, and the volume formulas for cylinders, cones, and spheres all serve one blueprint.

Standards Addressed

- 8.G.B.6 Explain the Pythagorean Theorem and its converse
- 8.G.B.7 Apply the Pythagorean Theorem in 2 and 3 dimensions
- 8.G.B.8 Distance between points
- 8.G.C.9 Volumes of cones, cylinders, and spheres

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with skate park photos. Part 1: ramps and rails.
Day 2	Part 2: map distances and layout.
Day 3	Part 3: volumes, concrete order, and blueprint.

Materials

- Student handout
- Grid paper
- Calculators with π
- Rulers
- Optional: cylinder, cone, and sphere models

How the Three Versions Differ

Level	What changes
CP2	Find the hypotenuse with starters, distances that form 6-8-10 triangles, and single volume formulas.
CP1	Missing legs, converse checks, coordinate distances, and unit conversion for a concrete order.
Honors	3D diagonals, scaled ramps, midpoint and perimeter, solving for a missing dimension, and a budget.

Watch For (Common Misconceptions)

- Adding the legs instead of their squares.
- Using the diameter instead of the radius in volume formulas.
- Dividing cubic feet by 3 instead of 27 to get cubic yards.

Extension Ideas

- Build a scale model of one ramp.
- Research real skate park ramp angles and compare them.
- Find the surface area of the bowl for a coating estimate.

Answer Key – CP2

Part 1: Ramps and Rails

#	Task	Answer
1	A ramp rises 5 feet over a run of 12 feet. How long is the ramp surface?	$5^2 + 12^2 = 169$, so c = 13 feet.
2	A rail goes 8 feet across and 6 feet up. How long is the rail?	$6^2 + 8^2 = 100$, so 10 feet.
3	Is a triangle with sides 7, 24, and 25 feet a right triangle? Explain.	Yes: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$.

Part 2: Map Distances

#	Task	Answer
4	The entrance is at (1, 2) and the half-pipe is at (7, 10). How far apart are they in grid units and in feet?	$\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ units = 50 feet.
5	Find the distance between the bench at (−2, 3) and the fountain at (4, −5) in grid units.	$\sqrt{(6^2 + 8^2)} = 10$ units.
6	The park is a 60-foot by 80-foot rectangle. How long is the diagonal path across it?	$\sqrt{(60^2 + 80^2)} = \sqrt{10{,}000} = 100$ feet.

Part 3: Concrete and Planters

#	Task	Answer
7	The bowl is a cylinder with radius 10 feet and depth 6 feet. Find its volume.	$V = \pi(10^2)(6) = 600\pi \approx 1{,}885.0$ ft^3.
8	Each planter is a cone with radius 3 feet and height 4 feet. Find its volume.	$V = \frac{1}{3}\pi(3^2)(4) = 12\pi \approx 37.7$ ft^3.
9	Decorative concrete balls are spheres with radius 1 foot. Find the volume of one ball.	$V = \frac{4}{3}\pi(1^3) \approx 4.2$ ft^3.

Reflection

#	Prompt	Look for
R1	How do you know which side is the hypotenuse?	It is the longest side, across from the right angle.
R2	How is the volume of a cone related to the volume of a cylinder with the same base and height?	The cone holds $\frac{1}{3}$ as much.

Answer Key – CP1

Part 1: Building Ramps

#	Task	Answer
1	A 10-foot ramp covers a base of 8 feet. How high does it rise?	$10^2 - 8^2 = 36$, so 6 feet.
2	A 15-foot rail reaches 9 feet high. How far along the ground does it reach?	$15^2 - 9^2 = 144$, so 12 feet.
3	A builder measures a frame with sides 9, 12, and 16 feet. Is the corner a right angle?	No: $9^2 + 12^2 = 225$, but $16^2 = 256$.

Part 2: Planning Paths

#	Task	Answer
4	Find the distance between (−3, 4) and (5, −2) in units and in feet.	$\sqrt{(8^2 + 6^2)} = 10$ units = 50 feet.
5	Is the triangle with vertices (0, 0), (6, 0), and (6, 8) a right triangle? How long is the longest side?	Yes: it has a horizontal and a vertical side. Longest side $\sqrt{(36 + 64)} = 10$ units.
6	How much shorter is the diagonal path across a 60-foot by 80-foot park than walking along two edges?	Diagonal 100 ft vs. 140 ft: 40 feet shorter.

Part 3: Ordering Concrete

#	Task	Answer
7	How many cubic yards of concrete would fill the bowl (radius 10 ft, depth 6 ft)? Round to the nearest tenth.	$600\pi \approx 1{,}885.0$ ft$^3 \div 27 \approx 69.8$ yd^3.
8	How much soil fills 6 cone planters (radius 3 ft, height 4 ft)?	$6 \times 12\pi = 72\pi \approx 226.2$ ft^3.
9	Twenty spherical balls of radius 1 foot are made from concrete. How many cubic feet are needed?	$20 \times \frac{4}{3}\pi \approx 83.8$ ft^3.

Reflection

#	Prompt	Look for
R1	How does the converse of the Pythagorean Theorem help a builder?	If $a^2 + b^2 = c^2$ for the sides, the corner is a right angle; if not, the frame is not square.
R2	Why must you convert cubic feet to cubic yards by dividing by 27, not 3?	A cubic yard is 3 ft × 3 ft × 3 ft = 27 ft^3.

Answer Key – Honors

Part 1: Ramps in 2D and 3D

#	Task	Answer
1	A storage box is 3 ft by 4 ft by 12 ft. How long is the longest pole that fits inside (corner to opposite corner)?	Floor diagonal $\sqrt{(9 + 16)} = 5$; space diagonal $\sqrt{(25 + 144)} = 13$ feet.
2	A ramp has the same steepness as the 5-over-12 ramp and rises 7.5 feet. Find its run and its length.	Scale factor 1.5: run 18 feet, length 19.5 feet.
3	A support has a hypotenuse of 10 feet and one leg of 7 feet. Find the other leg exactly and to the nearest hundredth.	$\sqrt{(100 - 49)} = \sqrt{51} \approx 7.14$ feet.

Part 2: Site Layout

#	Task	Answer
4	Find the distance between (−4, −1) and (8, 4).	$\sqrt{(12^2 + 5^2)} = 13$ units.
5	A bench goes halfway between those two points. Where is it?	$((-4 + 8) \div 2, (-1 + 4) \div 2) = (2, 1.5)$.
6	Find the perimeter of the triangular zone with vertices (0, 0), (9, 0), and (0, 12).	$9 + 12 + \sqrt{(81 + 144)} = 9 + 12 + 15 = 36$ units.

Part 3: Bowl Engineering

#	Task	Answer
7	A new bowl is a half-sphere with radius 9 feet. Find its volume.	$\frac{2}{3}\pi(9^3) = 486\pi \approx 1{,}526.8$ ft^3.
8	Compare it with a cylinder bowl of radius 9 feet and depth 9 feet. What fraction of the cylinder is the half-sphere?	Cylinder 729π ft^3; $\frac{486\pi}{729\pi} = \frac{2}{3}$.
9	A cone planter has radius 3 feet and the same volume as a sphere of radius 3 feet. How tall is it?	$\frac{1}{3}\pi(9)h = 36\pi \rightarrow h = 12$ feet.
10	How much would the concrete for the half-sphere bowl cost?	$1{,}526.8 \div 27 \approx 56.5$ yd^3 × \$150 ≈ \$8,482.

Reflection

#	Prompt	Look for
R1	How is finding a 3D diagonal like using the Pythagorean Theorem twice?	First find the diagonal of the base, then use it as a leg with the height.

#	Prompt	Look for
R2	Why is a half-sphere bowl $\frac{2}{3}$ of a cylinder with the same radius and depth?	Half-sphere: $\frac{2}{3}\pi r^3$. Cylinder with h = r: πr^3. The ratio is $\frac{2}{3}$.